Spacetime, propositional eternalism, and temporal egalitarianism

Handout for *Time and Modality* seminar

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1 A picture

The Manifold Picture: Concrete reality consists, fundamentally speaking, of *spacetime points* and *spacetime regions*. The structure of concrete reality is fully captured by a four-dimensional *differentiable structure* (a criterion for what counts as a "smooth" co-ordinate system) over the set of spacetime points, together with a short list of 'physically distinguished' scalar, vector and/or tensor fields (e.g. the metric field, the electromagnetic field, the mass-density field) on this differentiable structure.

2 Spelling it out

Manifold Supervenience: All true propositions[1] follow with metaphysical necessity from (i) truths about which spacetime points and regions there are; (ii) truths to the effect that a certain point or region is part of a certain other region; (iii) truths about which co-ordinate systems over the set of spacetime points are smooth; (iv) the truth, for each physically distinguished tensor field and spacetime point, about which tensor at that point is the value of that physically distinguished field; and (v) a "that's all" truth.

Manifold Naturalness: The only natural classes of points and regions are those picked out by simple conditions on the physically distinguished fields.

Manifold Homogeneity: None of the especially natural classes of spacetime points is a connected, spacelike, three-dimensional spacetime region.

Manifold Completeness: There is no way to embed the spacetime manifold with its physically distinguished fields into a larger connected manifold of the same dimensionality on which all the physically distinguished fields are everywhere smooth.

3 Two arguments

The Supervenience Argument:

(S1) The manifold picture is true.

(S2) If the manifold picture is true, then if any propositions are temporarily true, some propositions about the values of the physically distinguished fields at particular spacetime points are temporarily true.

(S3) If the manifold picture is true, then all true propositions about the values of the physically distinguished fields at particular spacetime points are eternally true.

Conclusion: There are no temporarily true propositions.[2]

The Homogeneity Argument:

H1 The manifold picture is true.

H2 If the manifold picture is true, then instants of time are connected, spacelike, three-dimensional regions of spacetime.

H3 If the manifold picture is true, then no connected, spacelike, three dimensional

[1]Or: all true *factual* propositions.

[2]An alternative version of this argument replaces 'proposition' with 'factual proposition' throughout.

region of spacetime has an especially natural unit set.

Conclusion: The unit set of the present time is not especially natural.

4 Ways in which the Manifold Picture is stronger than necessary for the argument

Some possible modifications that wouldn't matter:

- Properties and relations on the manifold that aren't naturally thought of in field-theoretic terms.
- Eliminate regions and keep points; or eliminate points and keep just some special regions.
- More radical changes to the ontology.
- Add particles or other 'occupiers' as additional fundamental ontology (so long as they don't break Homogeneity).

5 Response strategies

Radical responses: forget about the manifold altogether.

Moderate responses: subtract from the manifold.

Conservative responses: add to the manifold. The "Supplemented Manifold Picture": Manifold Supervenience + Manifold Naturalness + the denial of Manifold Homogeneity.

6 Relativity: the significance of Minkowskian structure

Three kinds of "flat" geometric structure on a 4d manifold.

- *Newtonian structure*: the geometry provides a natural way of picking out (i) a 'foliation', or partition of the 4d manifold into 3d hyperplanes (the 'hypersurfaces of simultaneity'), and (ii) a partition of the 4d manifold into 1d lines ('static trajectories').
- *Galilean structure*, a.k.a. Neo-Newtonian structure. The geometry still provides a natural foliation. But there is no longer a distinguished partition into lines; there is no geometrically definable notion of a line being "perpendicular" to a hypersurface of simultaneity. There is a distinguished class of 1d submanifolds, the 'unaccelerated trajectories'; but they are not a partition since they overlap. There is a natural equivalence relation of 'parallelism' between unaccelerated trajectories, such that parallel trajectories don't overlap.
- *Minkowskian structure*. The geometry does not provide a natural partition into 3d submanifolds or 1d submanifolds. There is a distinguished class of 3d submanifolds (the 'flat spacelike hypersurfaces') and a distinguished class of 1d subamnifolds (the 'unaccelerated trajectories'), such that each flat spacelike hypersurface intersects each unaccelerated trajectory at exactly one point. There are natural equivalence relations of parallelism among flat spacelike hypersurfaces and among unaccelerated trajectories. And there is a natural relation of **perpendicularity** between hypersurfaces and trajectories, such that all the unaccelerated trajectories perpendicular to some flat spacelike hypersurface are parallel, and vice versa.

Upshot: if we supplement Minkowski spacetime with some extra structure that provides a natural way of picking out a flat spacelike hypersurface (or family of such hypersurfaces), we will ipso facto have *also* provided a natural way of picking out a family of parallel unaccelerated trajectories—a 'criterion of motion and rest'. When supplemented with a natural flat spacelike hypersurface, Minkowskian geometric structure becomes as rich as (in fact, richer than) Newtonian structure.

7 The centre of mass of the universe

Of course there is more to reality than 4d geometry! If the non-geometric structure can be associated with something like a *mass-density field*, and the total mass of the universe (the result of integrating mass-density across any flat spacelike hypersurface) is finite, then the non-geometric structure will provide a natural way of singling out an unaccelerated trajectory: the trajectory of the centre of mass of the universe.

So be careful about saying things like 'In Minkowski spacetime, absolute simultaneity is not well-defined'. What's true is that there is no way of defining any non-trivial equivalence relation among spacetime points *just in terms of the geometric structure* (i.e. the differential structure and the metric field).

Suppose one embraces a Supplemented Minkowskian Manifold (Minkoskian manifold plus natural property of Presentness instantiated by points on a single flat spacelike hypersurface).

8 General relativity

In many (but not all) cosmologically realistic models of general-relativity, there *are* some natural foliations into spacelike hypersurfaces.

In some such models, there are *several* such natural foliations. (Example: constant mean curvature slices vs. slices of constant distance from the initial singularity).